

Shekoufeh Sadeghi

Ph.D. Candidate in Computer Science | Robotics, Active Perception, Computer Vision
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LinkedIn | GitHub | Google Scholar | [personal website / project page]

Research Focus

Active perception and perception-aware planning for robot autonomy; 3D vision and visual localization; learning and optimization for sensing under real-world constraints; robot learning and long-horizon manipulation. [Tailor this line per internship: field robotics vs. manipulation vs. CV/ML.]

Education

University at Buffalo, SUNY

Ph.D. in Computer Science; Advisor: Prof. Karthik Dantu

Buffalo, NY
2023 – Present

- Research in the DRONES Lab on active perception, robot vision, and perception-aware autonomy.
- [Expected graduation date]; [Ph.D. GPA only if you want it shown].

K.N. Toosi University of Technology

B.Sc. in Computer Engineering

Tehran, Iran
2017 – 2022

- Undergraduate research in computer vision, reinforcement learning, and generative models.

Research Experience

DRONES Lab, University at Buffalo

Graduate Researcher / Research Project Assistant

Buffalo, NY
2023 – Present

On-Manifold Active View Optimization for Motion-Decoupled Cameras

[2023/2024 – Present]

- Developed an online optimization method on $SO(3)$ for orienting motion-decoupled cameras to maximize task-relevant visual feature visibility while the robot follows a fixed translational trajectory.
- Built simulation and real-robot evaluation pipelines using ROS, C++/Python, COLMAP, Unreal/Isaac simulation, and a Boston Dynamics Spot equipped with a RealSense D455.
- Public project results report ~ 460 ms optimization time for ~ 15 k features versus up to 90 s brute-force search, and 99.04% feature-registration success versus 70.48% without optimized viewing. [Verify these numbers against the final paper before submitting].
- Extended perception-aware planning experiments to compare visibility optimization with map-based information objectives and localization baselines. [Add your strongest final quantitative localization result].

Adaptive Illumination Control for Robot Perception (Lightning)

2025 – 2026

- Co-authored a closed-loop illumination-control framework for visual SLAM that combines learned relighting, offline intensity scheduling, and imitation learning for online light control.
- [Replace with your specific contribution: e.g., data collection, CLID experiments, robot evaluation, SLAM evaluation, implementation, analysis].
- [Add one quantitative result that directly reflects your contribution, not only the paper's overall result].

Task-Aware Active Perception for Long-Horizon Manipulation (Ongoing)

2026 – Present

- Investigating view planning for manipulation that prioritizes task-relevant object-part visibility and action/skill feasibility rather than generic information gain.
- Exploring learned reachability / affordance models and backward or goal-conditioned planning to select useful intermediate sensing states for long-horizon tasks.
- [Add simulator/robot (e.g., UR5 + Robotiq 85), benchmark, concrete implementation status, and first result once available].

Institute for Research in Fundamental Sciences (IPM)

Research Intern – GameGAN Reimplementation and Improvement

Tehran, Iran
Jul 2021 – Dec 2021

- Reimplemented a GameGAN data pipeline and model; modified network/input architecture and debugged legacy dependencies, Gym environments, and CMake build issues.

Selected Publications

S. Sadeghi, Y. Turkar, and K. Dantu. “Adaptive Illumination Control for Robot Perception.” *arXiv preprint*, 2026.

D. Oliva, A. Olszewski, S. Sadeghi, K. Dantu, and D. Feil-Seifer. “Designing Socially Assistive Robots for Clinical Practice: Insights from an Asynchronous Remote Community of Speech-Language Pathologists.” *Frontiers in Robotics and AI*, 2025.

S. Milani, A. Momtaz, S. Amini, K. Amini, S. Sadeghi, E. Qoreishi, and S. Haddadian. “A Mathematical Approach Towards Random Road Profile Generation Based on Chaotic Signals of Chua’s Circuit.” *Contemporary Mathematics*, 2022.

S. Sadeghi. “Maximizing Feature Visibility by Online On-Manifold Optimization for Motion-Decoupled Cameras.” Northeast Robotics Colloquium (NERC) poster, 2024. [Add manuscript status / coauthors / venue if submitted].

Technical Skills

Programming: Python, C++, C, [CUDA if interview-ready]; Git, CMake, Linux, Docker

Robotics / Perception: ROS, ROS2, OpenCV, COLMAP, visual localization / SLAM, Eigen, Ceres, OMPL, RRT*, trajectory optimization

ML / Scientific Computing: PyTorch, TensorFlow, Keras, NumPy, SciPy, scikit-learn, Pandas, Matplotlib

Simulation / Hardware: NVIDIA Isaac Sim, Unreal Engine / UnrealCV, [Gazebo if current], Boston Dynamics Spot, Intel RealSense D455, [UR5 + Robotiq 85 if you have hands-on experience]

Other: LaTeX, [Jetson / Orin if you have deployed on it], [Open3D / PyTorch3D / JAX only if actually used]

Teaching & Mentoring

University at Buffalo

Teaching Assistant, CSE 4/568 Robotics Algorithms

Buffalo, NY

[semester(s)]

- [Add 1 strong line: ROS2/simulation assignments, office hours, grading, lab support, number of students].

K.N. Toosi University of Technology

Lead Teaching Assistant – Computer Vision; Embedded System Design

Tehran, Iran

2021 – 2022

- Led lab/workshop instruction, assignment design and grading, and coordination of teaching assistants.